

Dr. Khalil Abdulghani Alkebsi

Principal Engineer - Manufacturing Digitalization | AI & Robotics Leader | Autonomous Systems | Smart Manufacturing

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EXECUTIVE PROFILE

PhD-level AI, Robotics, Autonomous Systems, and Digital Transformation leader with 8+ years of industry and research experience across XPENG, BYD, and SANY. Currently Principal Engineer - Manufacturing Digitalization at XPENG, leading production-grade robotics and AI automation initiatives for intelligent automotive manufacturing. Specialized in autonomous robotic assembly, multi-robot coordination, motion planning, trajectory optimization, perception, AI-driven manufacturing digitalization, ROS/ROS2 systems, embodied AI, VLA robotics, and enterprise AI transformation. Strong record of translating research into deployable industrial systems, leading cross-functional teams, and building innovation platforms that combine robotics, AI, optimization, manufacturing execution, LLM/VLM-enabled intelligence, AI agents, RAG-based knowledge systems, and organizational digital transformation.

AI & Robotics Leadership	Industrial Deployment	Autonomous Systems	Government/Enterprise AI Fit
PhD + XPENG/BYD/SANY background; principal-level engineering leadership.	Production-ready automation, sub-millimeter assembly optimization, factory digitalization.	ROS2, MoveIt2, Nav2, Open-RMF, planning, control, VLA, embodied AI.	AI transformation, multi-agent systems, smart infrastructure, robotics strategy.

CORE LEADERSHIP & TECHNICAL STRENGTHS

- AI-driven manufacturing digitalization, Industry 4.0, smart factory transformation, autonomous production systems, and enterprise AI adoption.
- Robotics architecture: ROS1/ROS2, MoveIt2, Nav2, MoveIt Task Constructor, Open-RMF, multi-arm robots, AMRs, orchestration, and collision-free trajectory generation.
- Autonomous systems: perception, localization, planning, decision-making, control, trajectory optimization, calibration uncertainty handling, and deployment validation.
- Embodied AI and VLA robotics: vision-language-action concepts, multimodal robot reasoning, foundation-model-enabled robotic task execution, Pi0.5/RL-token direction, and industrial use cases.
- Leadership: cross-functional team coordination across robotics, perception, controls, software, mechanical integration, manufacturing engineering, production stakeholders, and enterprise/government AI transformation use cases.

PROFESSIONAL EXPERIENCE

XPENG Motors - Principal Engineer, Manufacturing Digitalization

2025 - Present | Guangzhou, China

- Lead AI, robotics, and manufacturing digitalization initiatives for advanced automotive production, connecting research, factory engineering, software, perception, control, and deployment teams around real industrial innovation.
- Invented and led development of X-Fit, an AI-powered autonomous door assembly and alignment innovation that transforms manual vehicle fitting into an intelligent robotic manufacturing capability.
- Built the technical architecture behind AI-guided 6D pose generation, gap/flush optimization, hinge and screw-hole alignment, and robust correction under sensor, calibration, and production uncertainty.
- Demonstrated production-oriented performance targets including approximately 0.1 mm alignment precision and sub-second optimization response in tested scenarios.
- Addressed real manufacturing challenges including measurement mismatch, structural variation, calibration drift, sensor noise, body/door tolerance differences, and deployment validation.
- Drove the shift from manual adjustment workflows toward intelligent, autonomous robotic fitting, reducing dependence on expert manual operations and enabling scalable smart manufacturing.
- Extended the innovation concept toward additional body-fitting applications including hood, back door, side window, corner panel, and other body-in-white assembly scenarios.

BYD - AI / Robotics / Autonomous Systems Engineering Roles

Previous Role | China

- Worked on intelligent automation, AI algorithms, robotics, and industrial engineering applications in automotive manufacturing environments, building practical experience in converting algorithms into factory-oriented solutions.
- Contributed to perception, planning, optimization, control, and deployment-focused engineering solutions supporting advanced manufacturing, autonomy, and production improvement.
- Built experience in large-scale industrial constraints, quality requirements, system integration, and cross-disciplinary execution across manufacturing, software, control, and operations teams.

SANY Heavy Industry - Autonomous Systems & Industrial Automation Roles

Previous Role | China

- Developed and supported autonomous systems and intelligent machinery applications for heavy equipment, connecting localization, guidance, control, sensing, operator interfaces, and field deployment.
- Contributed to the ICT excavator navigation and guidance product, integrating sensors, IMU/GPS positioning, controllers, HMI, wiring, and real-time visualization into a practical machine-mounted system.
- Delivered applied engineering solutions recognized by client stakeholders and aligned with real operating conditions, usability, and industrial value.

FLAGSHIP INNOVATIONS & TECHNOLOGY LEADERSHIP

X-Fit - Autonomous Robotic Door Assembly Platform

- Invented and led development of an autonomous, collision-aware robotic platform for precise vehicle door fitting and assembly in automotive production lines.
- Combines robotic manipulation, AI-guided 6D pose generation, sensor-driven measurement, optimization, and manufacturing constraints into one deployment-oriented system.
- Targets ≤ 0.1 mm gap/flush precision and fast optimization suitable for production environments.
- Solves practical uncertainties including calibration error, measurement inconsistency, sensor noise, structural deformation, and fitting deviation between door and vehicle body.
- Designed for extensibility from door fitting to hood, trunk/back door, side window, and other body-in-white assembly applications.

Otraj - Autonomous Collision-Free Trajectory Generation & Robotic Orchestration Platform

- Created Otraj as a code-free, multi-purpose robotics orchestration and collision-free trajectory generation platform based on ROS2, MoveIt2, Nav2, MoveIt Task Constructor, Open-RMF, and related robotics middleware.
- Provides autonomous task generation, robot coordination/orchestration, collision-free trajectory planning, and adaptable deployment across robotic arms and AMR systems.
- Simplifies complex robot programming and enables engineers, integrators, and operators to deploy scalable industrial robotic applications without requiring deep low-level robotics coding expertise.
- Supports multi-robot and multi-agent concepts for coordinated robotic workflows, industrial automation cells, and mobile manipulation systems.
- Strategic relevance: directly aligned with AI agents, robotics orchestration, autonomous operations, smart factory platforms, and embodied AI deployment.

VLA Dual-Arm + AMR Industrial Robotics Project

- Developing state-of-the-art Vision-Language-Action and embodied AI concepts for dual-arm robotic systems integrated with AMRs in automotive manufacturing contexts.
- Focuses on multimodal task interpretation, robot reasoning, autonomous execution, mobility, manipulation, and human-machine workflow integration, including current exploration of Pi0.5 and RL-token-based approaches.
- Connects foundation-model intelligence with real robotic constraints: perception, planning, grasping/manipulation, navigation, collision avoidance, safety, and deployment validation.
- Positions industrial robotics toward next-generation autonomous agents capable of understanding instructions, reasoning about tasks, and executing flexible physical operations in complex manufacturing environments.

Agricultural Excavator ICT Navigation & Guidance System

- Invented and led development of an intelligent excavator navigation and guidance product integrated directly into agricultural/heavy machinery.
- Integrated GNSS/GPS, IMUs, sensors, controllers, wiring, display screen, HMI, localization, navigation, visualization, and operator guidance into a field-deployable machine-mounted system.
- Enabled operators to visualize machine position, excavation direction, target points, and the surrounding operating environment in real time, improving precision, usability, and field productivity.

EDUCATION

Ph.D. in Control Theory and Engineering

Key Laboratory of Advanced Control and Optimization for Chemical Processes, East China University of Science and Technology, Shanghai, China | 2016/09 - 2020/12

Thesis: Fast Multi-/Many-objective Optimization and Its Application in Vacuum Pressure Swing Adsorption. Research focused on evolutionary algorithms, multi-objective and many-objective optimization, machine-learning-based optimization, and applications in CO2 capture using VPSA processes and catalyst design for CO2 conversion.

Master degree in Control Theory and Engineering

Key Laboratory of Advanced Control and Optimization for Chemical Processes, East China University of Science and Technology, Shanghai, China | 2013/09 - 2016/06

Research focused on model predictive control, optimization, dynamic chemical process control, single-objective optimization algorithms, and applications to distillation towers, reactors, and separation columns.

Bachelor degree in Automation Engineering

East China University of Science and Technology, Shanghai, China | 2009/09 - 2013/06

RESEARCH, PUBLICATIONS & ACADEMIC PROFILE

- Author of multiple SCI-indexed publications related to AI, optimization, robotics, control systems, autonomous systems, and intelligent manufacturing.
- Google Scholar profile available with citation record and publication list: <https://scholar.google.com/citations?user=IKBAQwYAAAAJ&hl=ar>
- Book chapter contribution in optimization and machine learning: surrogate model-based optimization, evolutionary algorithms, black-box modeling, and real-process validation.
- Research interests: embodied AI, robotic foundation models, trajectory planning, autonomous decision-making, smart manufacturing, multi-agent robotics, and AI transformation.

SELECTED TECHNICAL KEYWORDS

AI for Manufacturing | Robotics | Autonomous Systems | Smart Manufacturing | Industry 4.0 | Manufacturing Digitalization | Enterprise AI | LLMs | VLMs | RAG | AI Agents | Multi-Agent Systems | Embodied AI | Vision-Language-Action (VLA) | ROS | ROS2 | MoveIt2 | Nav2 | MoveIt Task Constructor | Open-RMF | Motion Planning | Collision-Free Trajectory Generation | Multi-Robot Coordination | AMR | Dual-Arm Robotics | Perception | Control | Optimization | 6D Pose Estimation | Calibration | Sensor Fusion | Digital Transformation | Enterprise AI Architecture | Production Deployment

EXECUTIVE POSITIONING FOR UAE / GOVERNMENT AI OPPORTUNITIES

- Can support national AI and robotics programs by translating advanced AI research into deployable industrial, infrastructure, smart government, and intelligent machinery systems.
- Strong fit for government-backed AI, robotics, smart mobility, smart manufacturing, sovereign AI, industrial automation, autonomous operations, enterprise AI, and digital transformation programs.
- Combines rare experience across top Chinese EV/manufacturing groups, academic research, and real industrial deployment - valuable for organizations seeking practical AI transformation rather than only software experimentation.
- Target roles: Principal AI & Robotics Architect, Head of Industrial AI, Director of Robotics & Autonomous Systems, Manufacturing Digitalization Lead, AI Transformation Lead, Autonomous Systems Lead.

AWARDS & RECOGNITION

- 2nd Place (Runner-Up) — XPENG AI Open Competition: awarded among 6,813 teams worldwide for X-Fit, the AI-powered autonomous car-door alignment and fitting system achieving up to 0.1 mm gap/flush precision for smart manufacturing.
- Certified Inventor of X-Fit — XPENG Certificate of Achievement recognizing the invention of the X-Fit next-generation intelligent robotic door-fitting technology.